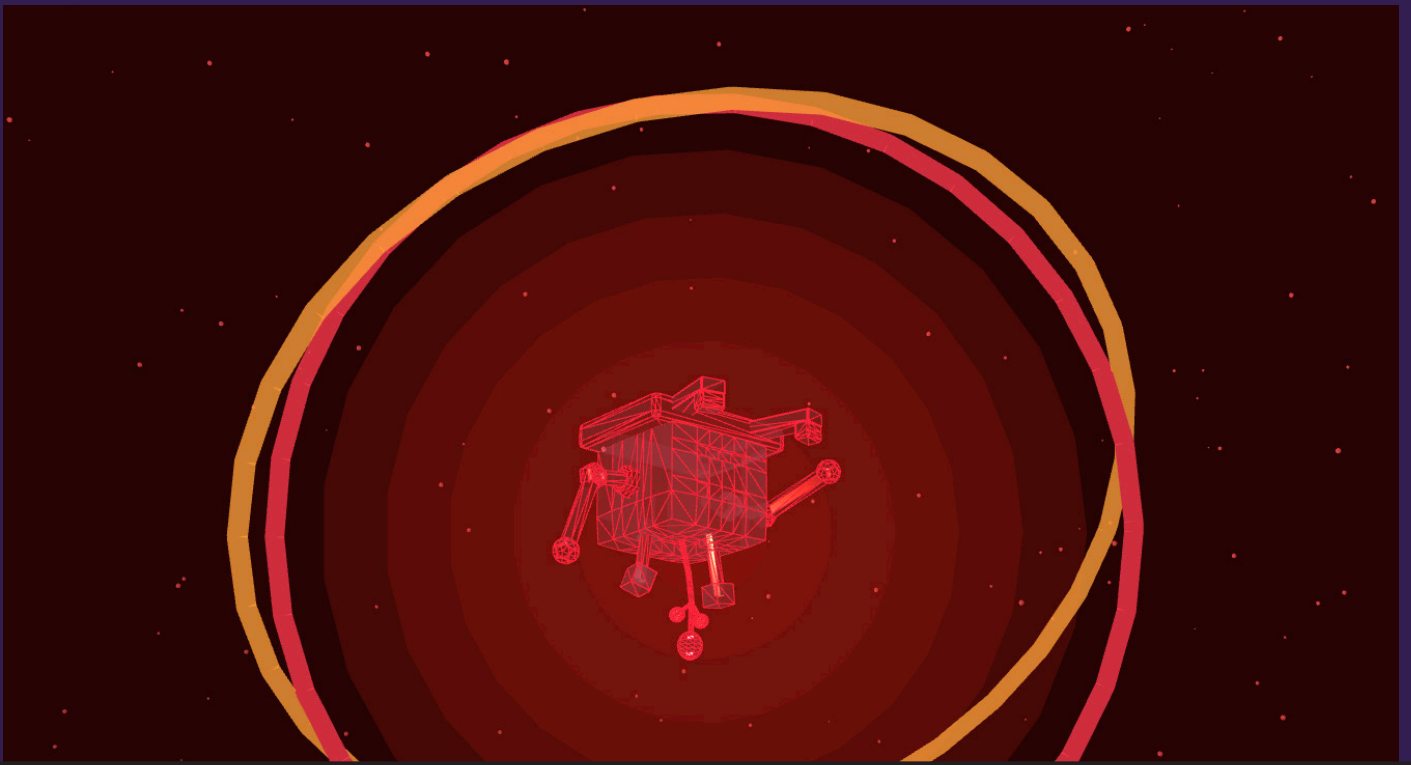


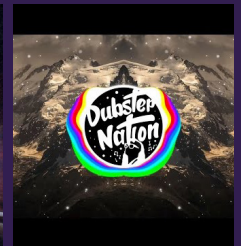
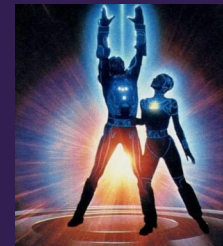
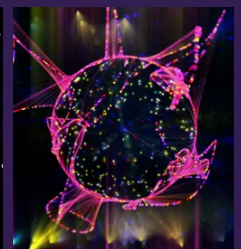
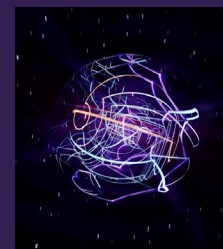


Creative coding - AudioVisual project



Robot Space waves

For my Project I wanted to incorporate vintage toy design with 80s and 2000's vibes that'd give my project a retro feel that has been adapted to work well with modern sounds.



For examples I've included things I have taken inspiration from for my original robot model and also things I wanted to incorporate. Heavy inspiration came from the old dubstep nation reactive videos

Where to start?

We have previously learnt about using microphone and sound inputs to be able to make an image that will change and react to bass and treble of a song using the values to determine what to do with the image ie. Making a ring that gets bigger / pulses on bass drops and changes making it interactive with the song and also making it completely unique to every song made. With the inspiration I already Had I wanted to adapt my Learning to make something new, We also had learned previously how to use WebGL To draw 3d objects on java script, I discovered from further research that you are able to load 3d models using webgl.

Coding

```
✓ function preload() {  
  | robot = loadModel('robot.obj', true, handleModel, handleError);  
  | }  
}
```

This is the Core fundamental behind My code being to load the model before anything else This combined with the units for bass and treble I am able to use the preload to change it for these values.

```
if (loaded && sound.isPlaying()) {  
  | fft.analyze();  
  | bassN    = fft.getEnergy('bass')      / 255;  
  | midN     = fft.getEnergy('mid')       / 255;  
  | trebleN  = fft.getEnergy('treble')    / 255;  
  | level   = (bassN + midN + trebleN) / 3;  
}
```

Layers Of Interactivity

```
// ----- REACTIVE BACKGROUND -----  
if (isRed) {  
  | // Hot red / crimson theme  
  | let bgR = map(bassN, 0, 1, 40, 200);  
  | let bgG = map(midN, 0, 1, 5, 30);  
  | let bgB = map(trebleN, 0, 1, 5, 40);  
  | background(bgR, bgG, bgB);  
} else {  
  | let bgR = map(bassN, 0, 1, 10, 120);  
  | let bgG = map(midN, 0, 1, 10, 60);  
  | let bgB = map(trebleN, 0, 1, 25, 180);  
  | background(bgR, bgG, bgB);  
}
```

```
//pulsing "glow" behind everything
push();
translate(0, 0, -900);
noStroke();
let glowSize = map(level, 0, 1, 1200, 2400);
for (let i = 6; i > 0; i--) {
  if (isRed) {
    fill(
      map(bassN, 0, 1, 180, 255),
      map(midN, 0, 1, 30, 120),
      map(trebleN, 0, 1, 20, 90),
      20 + i * 6
    );
  } else {
    fill(
      map(trebleN, 0, 1, 80, 255),
      map(midN, 0, 1, 40, 180),
      map(bassN, 0, 1, 120, 255),
      20 + i * 6
    );
  }
}
```

```
// ----- ROBOT -----
push();

// Spin speed driven by treble
let spinSpeed = map(trebleN, 0, 1, 0.01, 0.12);
angle += spinSpeed + 0.01;

rotateX(angle * 0.4);
rotateY(angle);

// Scale pulse on bass - base size doubled
let scaleFactor = map(bassN, 0, 1, 1.6, 3.6);
scale(scaleFactor);

// Shake on strong bass hit
if (bassN > 0.75) {
  translate(random(-6, 6), random(-6, 6), 0);
}
```

```
// Star/particle field reacting to treble
push();
noStroke();
for (let s of bgStars) {
  let tw = 0.5 + trebleN * 2.5;
  let sz = s.baseSize * tw;
  push();
  translate(s.x, s.y, s.z);
  if (isRed) {
    fill(
      255,
      80 + midN * 100,
      80 + trebleN * 60,
      150 + bassN * 105
    );
  } else {
    fill(
      200 + trebleN * 55,
      180 + midN * 75,
      255,
      150 + bassN * 105
    );
  }
  sphere(sz, 6, 6);
  pop();
  s.x += sin(frameCount * 0.005 + s.bue) * 0.3;
```

Different States

```
// ----- LIGHTS -----  
ambientLight(map(bassN, 0, 1, 30, 90));  
  
if (isRed) {  
  // Warm key light – red to orange with treble  
  let rR = map(trebleN, 0, 1, 200, 255);  
  let rG = map(trebleN, 0, 1, 40, 140);  
  directionalLight(rR, rG, 40, 0, -1, -1);  
} else {  
  let r = map(trebleN, 0, 1, 120, 255);  
  let b = map(trebleN, 0, 1, 255, 120);  
  directionalLight(r, 140, b, 0, -1, -1);  
}
```

To create different states I gave my model different modes being between red and green presenting different moods that can be changed for the song chose giving a more perosnal feel to each one and a layer of customiseability I would like to expand further on this idea maybe using the sound Values to assign the colours automatically and give even more options !

I knew I wanted to be able for users to choose there own songs. I did create the aesthetic with the songs I have chosen within mind but I really wanted it to be adaptable for potential dj set visuals asl believe they could add a great sound to visual element

```
function setup() {  
  createCanvas(1000, 1000, WEBGL);  
  
  fft = new p5.FFT(0.85, 1024);  
  
  // Create a file input button on the page  
  let btn = createFileInput(handleFile);  
  btn.position(20, 20);  
  btn.attribute('accept', 'audio/*');  
  
  // Play/pause button  
  let playBtn = createButton('Play / Pause');  
  playBtn.position(160, 20);  
  playBtn.mousePressed(togglePlay);  
  
  // Outline color toggle button  
  outlineBtn = createButton('Outline: Green');  
  outlineBtn.position(270, 20);  
  outlineBtn.mousePressed(toggleOutlineColor);  
}
```

References

Web GL and Bits

introduction - <https://www.youtube.com/watch?v=nqiKlWXUX-o8>

geometries - <https://www.youtube.com/watch?v=6TPVoB4uQCU>

lights and materials - <https://www.youtube.com/watch?v=k2FguXvqp60>

camera and perspective - <https://www.youtube.com/watch?v=BU3D9WwaIQE>

loading obj - <https://www.youtube.com/watch?v=FUI7HEEz9B0&t=277s>

File input - <https://www.youtube.com/watch?v=ScUjx4aWRi0>
<https://stackoverflow.com/questions/43710173/upload-and-play-audio-file-js>

FFT - <https://www.youtube.com/watch?v=203nm0flvbi4>

Play Button - <https://stackoverflow.com/questions/27368778/how-to-toggle-audio-play-pause-with-one-button-or-link>

Evaluation

Overall with this project I have been able to delve into what is possible with my coding, using visual elements and creativity in my code gave me a real feeling of passion for what I was making as it gave a more art sense to what I was making. I would love to delve back into this idea with different ideas in mind that I could include but I also want to see what other states could be included in this project as I believe that I can use this to my full potential when I think about the other things that the sound information can effect about my project such as the automatically chosen colours used depending on tempo perhaps? or maybe the BPM of the song ! It is infinitely tinkerable and I will be making sure to see what else is possible with it ! When I have the chance to use this model of code again I'd love to delve on different 3d states as well which I think could of worked really well to show the fft values